

AI Palmistry & Tarot Intelligence Platform

Milestone 3 — AI Interpretation Engine & Personalized Dashboard, Verification Report

Covers: `ai_interpretation_engine.py`, `dashboard_report.py`, `milestone3_pipeline.py`

Executive Summary

This milestone wires an LLM into the palm and tarot data verified in Milestone 2, turning grounded facts into a written interpretation, and packages the result into a personalized PDF dashboard. Each piece was verified independently before being trusted together, then run as one full pipeline end to end. All three passed on real data, not placeholders.

1. AI Interpretation Engine — Verification

Ran the module's self-test — no live LLM call, no API key needed — to confirm the prompt-building and JSON-parsing logic work correctly on their own before spending an API call on them.

Check	Result
Module imports cleanly and self-test runs	Yes — prompt built and printed correctly from a mock combined_reading
Prompt grounds itself in real data only	Confirmed — prompt explicitly instructs “Base your entire analysis ONLY on the factual data below,” and includes the actual tarot_text and palm findings, not placeholder text
JSON extraction handles Gemini's markdown fences	Confirmed — self-test fed a fenced ```json``` block; parser correctly stripped it and returned valid JSON matching the requested schema
Fallback path returns a complete, safe result on failure	Confirmed by design — a failed LLM call returns the same JSON keys with “Not available.” placeholders instead of raising and crashing the caller

Finding: the engine separates “getting real data into the prompt” from “trusting whatever comes back” — the JSON parser was tested against the messy, fenced format Gemini actually returns, not an idealized clean response.

2. Dashboard PDF & Full Pipeline — Verification

The real end-to-end test: Milestone 2's combined reading, fed through the interpretation engine, fed through the dashboard generator, run as one pipeline via `run_full_pipeline()`.

Step 1/3: Generating palm + tarot reading (Milestone 2)... Palm success: True Tarot source: llm Step 2/3: Running AI interpretation engine (Milestone 3)... Analysis source: llm Step 3/3: Building personalized dashboard PDF... Dashboard saved to: dashboard_report.pdf	
Check	Result
Combined reading feeds cleanly into the AI engine	Yes — no manual reshaping needed between <code>generate_combined_reading()</code> output and <code>generate_full_analysis()</code> input
AI interpretation used real (not fallback) analysis	Yes — <code>analysis["_source"] == "llm"</code> for this run
Dashboard built from both real data and AI analysis	Yes — all 5 sections populated: Source Data, Interpretation, Personality Intelligence, Recommendations, Life Trend Analysis
Fallback-shaped analysis also renders cleanly	Yes — <code>dashboard_report.py</code> 's self-test built a PDF from an all-“Not available.” analysis with no errors

Figure 1 — actual `dashboard_report.pdf` produced by this pipeline run:

Personalized Reading Dashboard

Generated: 2026-07-31T00:08:05
Analysis source: llm

Source Data

Question asked: What should I focus on this month?

Heart Line: Your Heart line is short, which means you will meet various people and have a broad range of relationships throughout your life.

Head Line: Your Head line is long, which means you will explore a broad range of topics throughout your life.

Life Line: Your Life line is short, which means you are independent and autonomous.

King of Swords (Upright) - Expressing yourself with firmness and authority; Rendering a final decision

Ace of Wands (Upright) - Being inspired; Identifying an important goal

The Emperor (Upright) - Exercising authority; Defining limits

Interpretation

Having recently cleared your path by expressing yourself with firmness and authority, your main focus this month is to use your vision to identify an important goal for creation. As an independent and autonomous individual, you are well-positioned to explore broad topics and channel your inspiration effectively. By embracing your inner genius and defining clear limits, you will bring strong organization and structure to this new direction.

Personality Intelligence

Strengths: Independence and autonomy, Ability to express yourself with firmness, authority, and expertise

Weaknesses: Need to actively establish structure and clear regulation, Requirement to define limits around expansive creative sparks

You act with independence and explore a broad range of topics and relationships throughout your life. When deciding on a path, you express yourself with firmness and expertise to clear the way forward.

What this shows:

the actual dashboard_report.pdf produced by this pipeline — real detected palm line findings (Heart/Head/Life), three real tarot cards, and an LLM-generated interpretation, personality read, recommendations, and life trend analysis, all grounded in the same session's real data, not templated filler.

3. Key Findings & Next Steps

Confirmed working

- Interpretation engine: real prompt grounding, tested JSON parsing, safe fallback shape
- Dashboard generator: builds correctly from both real and fallback analysis
- Full pipeline: one function call, three stages, real LLM analysis end to end
- Dashboard PDF: all five sections populated with real session data

Known limitations, honestly stated

- Fallback path is silent by design — a failed LLM call gives no visibility into why (bad key, network, malformed response) beyond inspecting _source directly
- JSON extraction relies on one regex for fence-stripping; unusual fence formatting could still fail to parse
- Dashboard's "Source Data" falls back to a generic sentence when palm_text is empty — exercised in testing, not yet common in real runs
- Verified with one demo image and one tarot draw; broader variability not yet stress-tested

Next step — Deployment: these three verified pieces are what Milestone 4 wraps into a tested FastAPI backend and a live Streamlit interface.

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